Paradigm Shifts: Motivated Inference on Webs of Belief

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Abstract. Paradigm shifts are a significant driver of belief revision in scientific communities. Peripheral commitments drift gradually under accumulating evidence, while central commitments resist revision even when evidence is against the theory—and when they do change, they change quickly. We study this phenomenon within an active inference framework, building on recent accounts that recast belief updating as motivated inference subject to a cost of changing one's mind. Drawing on Lakatos and Kuhn, we represent an agent's theoretical commitments as a directed network of interdependent beliefs, in which the cost of revising a commitment depends both on standing affective commitments and on how much surrounding inference must re-equilibrate when it moves—so the hard core and protective belt fall out of the precision geometry rather than being decreed. Agents exchange beliefs, and pieces of structure, over a communication network. Characterising the resulting dynamics analytically and through simulation, we find that the field's default aggregation step—precision-weighted averaging—cannot reproduce the signature of paradigm change: it contracts the community onto a compromise network no member holds, whatever the parameters. We then construct the representational capacity the default lacks—a posterior over rival structures within each agent—and show that it recovers the Kuhn cycle, with crisis as a measurable state of the belief: normal science, crisis, revolution, new normal. We close by discussing implications for engineered communities of reasoning agents, whose aggregation protocols decide the same question by design.

Keywords: Active inference · Structure learning · Belief sharing · Paradigm change · Social epistemology

1 Introduction: changing a frame, not a credence

If we want science to keep going well, we need to know how it *works*: what kind of agent can do science at all, how a community comes to represent a new dimension of the world, and how it loses information it once held. The question has a second audience, because communities of reasoning agents are now *built* as well as studied, and every such community commits, in its aggregation step, to an answer it rarely examines: what happens to the member who disagrees. Machine learning names the property at stake *open-endedness* [14, 28]; social epistemology catalogues its failure modes [23, 32]. What neither supplies is a

foundation: a model of a *community* of structure-learning agents explicit enough that one can prove which aggregation dynamics preserve open-endedness and which foreclose it. This paper tries to be a step towards that foundation. We show that the most natural baseline—Gaussian beliefs under precision-weighted pooling—provably lacks the capacity, and we construct a way for it to have that capacity. The problem, throughout, is averaging: taking the mean refines the frame everyone shares, and can never grow a new one.

To change one's mind about a fact is to update a belief; to change a paradigm is to change the *frame* in which beliefs are posed—not a credence but a *structure*, a network of conditional dependencies through which information propagates. This is the phenomenon Kuhn named but could not formalise [19]: nothing in *The Structure of Scientific Revolutions* says, mechanistically, why a community holds a structure of commitments past the first disconfirming evidence, nor what finally makes the holding give way. Take an example: when the chemical community of the late eighteenth century gave up phlogiston, it did not revise a single credence: it rewrote a network in which combustion, calcination, respiration, and the theory of acids were all load-bearing on a hub that had to go [40, 42, 43]. Phlogiston was load-bearing for too much.

We can model such communities with network epistemology, the dominant formalism for them—but in its naive form, a scalar credence in a fixed proposition updated on a trust graph [16, 24, 32, 45, 75], it cannot represent structure learning. The phlogiston puzzle is not a failure to converge on a number—the commitments were *entangled*, and the dispute was over which entanglements to keep. Thagard's ECHO [30, 41] captures the structural intuition as undirected coherence but lacks the directed dependencies and the carry-over cost of mind-change derived below; active-inference accounts supply precision-based lock-in [1], warnings that naive aggregation degrades collective precision [47, 48], and a single-agent variational cost of mind-change [15]. What none of these does is make *belief structure itself* the object of inference: a scalar credence can never say how revising one commitment forces a re-reading of the rest—agents can herd on a number or split over it, but they cannot come to hold different *kinds* of world.

This paper lets a piece of *structure* travel, so that one agent can come to represent a commitment its peers do not. Its agents carry a directed network of commitments and learn that network by Bayesian model *reduction*—and by reduction only: expansion is deferred by scope (§3.2), but when agents are *given* new structures, they can incorporate them into their own system. They are *bounded*, sampling a complex world through narrow channels, valuing some conclusions over others, and judging what they receive against their own model. Section 3 assembles this community from the field's own default choices; Section 4 shows what those defaults force—pooling is a contraction, so a conflicting source is never rejected, only averaged in—and what undoes it: agents that represent rival structures keep different readings of the same world apart, and run the Kuhn cycle with crisis as a state of the belief. The discussion draws the map: what filling the representational slot bought, and what remains.

2 Desiderata for a structural model of science

A structural model of science needs at least three ingredients. First, *structure as the object of inference*: a scientist confronting a new domain is not given the right variables, let alone the dependencies among them; frames differ in kind, not in degree, so credences over a fixed menu cannot carry them [34, 35, 49]. Active inference can model this: *when* structure is the object of inference, exploration need not be added by hand, as the ϵ of an ϵ-greedy algorithm is [24, 32]; it can be cast as the epistemic value of *expanding* the model to entertain a commitment not yet in it [9, 10], with Bayesian model reduction as its complement, pruning what the data abandon in closed form [11, 50]. To model structure learning, then, is to model how a Bayes net learns to change *itself* over time.

Value, or motivated belief updating: the underlying values and biases a scientist already has about what they care about. A result on which a career rests is held more tightly than its evidence warrants [17, 29], and attitudes demonstrably live on networks of exactly this kind [3]. And *communication*, because science is social: beliefs travel a trust graph and must somehow be combined when they meet. Each requirement has a default formalization, so natural it is hardly ever examined. Beliefs over a network become a precision-weighted Gaussian model; value becomes a utility that tilts the prior [15]; combination becomes precision-weighted averaging, the information form of classical opinion pooling [60, 62]. Section 3 builds the community out of exactly these defaults, on purpose. The question the rest of the paper asks is whether such a community can ever do more than *refine*: whether agents built this way can grow a frame, and not merely polish the one they hold.

3 The model

We draw the system whole first—a population of agents on a trust graph—then build the single agent up from first principles, and finish by stating the dynamics as an explicit per-round loop (§3.4).

3.1 The population, and the paradigm network each agent carries

We turn to the third requirement, communication, first. A population of N agents inhabits a shared hidden world. Agents are coupled on a trust graph: communities are blocks of a stochastic block model, with dense intra-community edges and an inter-community coupling weight (*inter*) that we sweep from total isolation to dense contact. Each agent privately carries its own *paradigm network*, a directed graph $G = (V, E, \Pi)$: nodes carry theoretical commitments x_v, edges carry inferential dependencies, and each edge bears a coupling precision π_{uv} measuring how tightly v is bound to u. The network is just a generative model whose hidden states are the commitments themselves: internal coherence is the precision-weighted graphical model $p(x \mid G) \propto \prod_{(u \to v) \in E} \psi_{uv}(x_u, x_v)^{\pi_{uv}}$, from

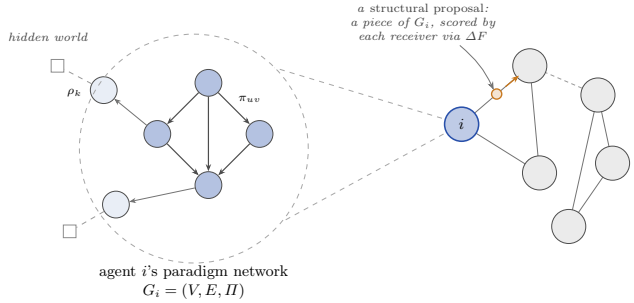


Fig. 1: The population model. Right: a trust graph, fixed at the start (stochastic block model)—dense channels within a community, one weak weight *inter* between. Left, magnified: each agent's private paradigm network $G_i = (V, E, \Pi)$, its belt reading the hidden world through channels of precision ρ_k. Along trust channels travel fused posteriors and *structural proposals*: pieces of paradigm each receiver scores against its own net by ΔF (Eq. 5).

which the agent's conservatism and conviction will be derived (Eqs. 1 and 2 below). Figure 1 draws the picture.

Two kinds of things propagate on this underlying network. Beliefs and accumulated precisions are fused across trust edges in information form—an agent's own evidence and its neighbours' weighted in proportion to trust, the information-form counterpart of trust-weighted averaging in opinion dynamics [55, 60]. And—not standard—*structure* spreads the same way: a neighbour's newly awakened node arrives as a proposal, scored against the receiver's own model (§3.2)—not only a number but a piece of structure, evaluated against wiring the sender does not see.

To see what one of those private networks holds, let us take an example. The phlogiston paradigm, at its smallest, is three commitments wired together: combustion releases phlogiston; calcination is combustion, slowed; respiration is combustion, tamed. Each commitment conditions the next, evidence for any one bears on all, and the question "what if the calx *gains* weight?" has no local answer—it tugs the whole web. A paradigm, in this paper, is that web made explicit.

Concretely, $p(x \mid G)$ is a linear-Gaussian structural equation model in the lineage of Wright's path analysis [74], edges as path coefficients and precisions as their weights; we work throughout in that regime, which puts every quantity below in closed form (Appendix A). Two stipulations: agents are *honest*—no one misreports, so lock-in is never a dishonesty artifact—and *bounded*, minimizing variational free energy step by step [39]—that is, they do not plan over longer horizons; they are myopic.

What we model is that revising a commitment is not a local act: changing x_v forces everything downstream of it to re-equilibrate, along every chain of dependence. If A is the matrix of edge precisions, summing chains of every length

gives $\sum_k A^k = (I - A)^{-1}$, a finite sum on a DAG (an object path analysis and network science already own [58, 59, 64, 70, 74]). We adopt it as the *propagation operator* $\mathsf{T} = (I - A)^{-1}$: the entry $\mathsf{T}_{iw} = \pi_{i \rightsquigarrow w}$ is the total precision-weighted coupling propagated from i to w along all directed paths (Appendix A).

Conservatism is then recognized rather than posited. Place a unit of weight on every node and ask, for each commitment i, how much total weight a revision of i would disturb. That is the operator applied to the unit source,

$$\kappa \;=\; \mathsf{T}\,\mathbf{1}, \qquad \kappa_i \;=\; 1 + \sum_{w \in \mathrm{desc}(i)} \pi_{i \rightsquigarrow w}, \tag{1}$$

the precision-weighted mass of a node's descendants. In network-science terms this is a centrality (the broadcast Katz index), usually read as influence or status; we read the same number as *revision cost*—a commitment is exactly as expensive to revise as the mass that must re-equilibrate when it moves. Conservatism is a centrality the graph already determines, not a disposition assigned by hand; one can read κ_i almost as a momentum or inertia term: the older and more central a commitment, the more mass moves with it.

What we add to this inherited machine comes in two layers. The first is contact with the world. The dense, high-precision interior of the network is the *core*; the sparse periphery in contact with observation is the *belt*, each belt node reading the world through a sensory channel of precision ρ_k—Lakatos's hard core and protective belt [66], here determined by the precision geometry rather than decreed. The central nodes are coupled to many more nodes; in our model, these are the *axioms* of the scientific system. The core carries no direct sensory channel: an interior commitment is revised only by error propagated inward from the belt, through the same operator—costly by Eq. (1) and insulated by position, which is what makes it slow.

The second layer is value, and the operator is run twice. This is the second desideratum of §2—motivated belief updating: scientists hold values and biases about what they care about, and a model of science has to represent them. We attach to each node an intrinsic utility u_i and let it propagate through the *same* operator: a commitment one wants true lends value to its dependents, so the conviction field is

$$U \;=\; \mathsf{T}\,u, \qquad U_i \;=\; u_i + \sum_{w \in \mathrm{desc}(i)} \pi_{i \rightsquigarrow w}\, u_w. \tag{2}$$

In the operator's path-analytic reading, U is the total effect of wanting things true. Where the inherited models propagate a single source, a paradigm carries two: $\mathsf{T}\mathbf{1}$ and $\mathsf{T}u$ are linearly independent unless $u \propto \mathbf{1}$, so the two fields need not align—a commitment can be central yet value-neutral, cheap yet strongly held (Fig. 5, Appendix A).

3.2 Motivated revision, structure learning, and forgetting

The bounded agent revises its belief $q(x)$ over the whole network by a motivated trade [15]: the variational cost of changing its mind—divergence from the prior

its own Bayes net encodes—against accuracy to the evidence and the value of where the commitments land,

$$\begin{aligned} \mathcal{F}[q] \;&=\; \underbrace{D_{\mathrm{KL}}[q(x)\,\|\,p(x\mid G)]}_{\text{mind-change cost}} \;-\; \underbrace{\mathbb{E}_q[\log p(o\mid x,G)]}_{\text{accuracy}} \;-\; \lambda\, \underbrace{\mathbb{E}_q\big[U^\top x\big]}_{\text{value}} \\ \Longrightarrow \qquad & q_\lambda(x) \;\propto\; p(x\mid o,G)\, e^{\lambda\, U^\top x}, \end{aligned} \tag{3}$$

with λ a single temperature governing how far value may bend the evidence. The KL is anchored to the *network prior*: mind-change is priced from where the mind currently stands, in the metric of the paradigm's own precision (Appendix A, Eq. 8), and the minimizer is classical—the exponential tilt KL-regularized utility maximization always yields [72, 73]. What is ours is the source of the tilt, a conviction field propagated through the paradigm's own operator. In the linear-Gaussian regime the tilt is a pure mean shift: value displaces where the commitments come to rest (by $\lambda\Sigma U$) and leaves the uncertainty untouched. What opposes the displacement is evidence. Evidence pulls a belief back through whichever sensory channels bear on it, and the strength of that pull scales with how much the agent *listens* to the channel—its sensory precision ρ_k. As conviction drives a channel's precision toward zero the pull vanishes, and the value-displaced commitment stands unrefuted. The fraction of disconfirming sensory precision so silenced,

$$\gamma \;=\; \frac{\sum_{k\in\mathrm{dis}} \rho_k \nVdash[\,\rho_k \text{ gated by the core}\,]}{\sum_{k\in\mathrm{dis}} \rho_k}, \tag{4}$$

is the *lock-in parameter*. Each round, channel k carries weight $w_k = \exp(-g\,|U\cdot H_k|)$, where H_k reads out which commitments the channel bears on: the more directly a channel bears on valued commitments, the less the agent listens to it, with g setting how strongly caring becomes not-listening. The weight is a per-channel, per-round quantity one can track over time—how much an agent is still listening, belief by belief in its Bayes net.

We now introduce structure learning. Structure learning revises G itself—the first requirement of §2: which dependencies survive in the network is itself the object of inference. Two asymmetric operations exist [10, 26], the formal counterpart of normal versus revolutionary science; *the model in this paper runs only reduction. Reduction* is the exact half: each edge carries an automatic-relevance prior precision [20, 31], and Bayesian model reduction [8, 11] scores any pruning in closed form from the already-inverted posterior,

$$\frac{p(o\mid \tilde{M})}{p(o\mid M)} = \mathbb{E}_{p(\theta\mid o)}\left[\frac{\tilde{p}(\theta)}{p(\theta)}\right], \tag{5}$$

the Savage–Dickey ratio in general form: one inversion scores every candidate reduced structure without re-touching the data. Conviction enters here too: a candidate prune is scored by its evidence ΔF plus λ times its conviction-value change ΔU (the loop of §3.4).

There is also *expansion*—entertaining a commitment no one has represented [10]—and it is outside the scope of this paper: it raises Stanford's problem of unconceived alternatives [27], and it would pull focus from the core point, representing divergence *within* a community. Where an experiment needs novelty it is supplied by hand: the rival candidates of §3.3 are given, not discovered.

The question that drives the rest of the model is *diversity*: how can a community of reduction-driven agents represent different readings of the same world—how can Bayesian model reduction do more than collapse onto mean approximations? The first requirement is *forgetting*: without it, accumulated precision in the Bayes net only grows, and nothing can ever change. Accumulated precision therefore relaxes toward the prior, $\Pi \leftarrow \Pi_0 + \omega(\Pi - \Pi_0)$, $\omega \in (0, 1]$ [36, 38]—one forgets at the rate one believes the world's structure is moving [37]. Without forgetting $\kappa = \mathsf{T1}$ can only grow and lock-in is automatic rather than earned; with it, the value tilt saturates at $\lambda U/(1-\omega)$ against also-saturated per-step evidence, so conviction holds only above a threshold it must clear.

3.3 Representing diversity: weights over observations and structures

What does representing diversity require? Concretely: a model in which A causes C directly is a different reading of the world from one in which A acts through B, and a community must be able to hold the two apart—not average them into a compromise that is neither. Two kinds of weight, both held constant in the defaults above, supply that capacity (Fig. 2).

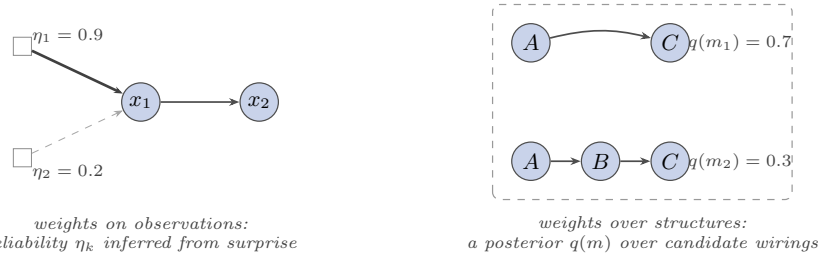


Fig. 2: Representing diversity. *Left:* each observation channel carries an inferred reliability weight η_k (Eq. 6)—a hyperprior through which trust is learned rather than fixed. *Right:* the same move one level up: an agent holds two candidate structures—$A \to C$ direct versus $A \to B \to C$—with a posterior $q(m)$ over them, so different readings of the same evidence are held apart rather than averaged.

Weights on observations. Stop treating a channel's precision as a constant and *infer* it [6]: carried as a scale mixture, each channel's noise precision becomes a latent with its own prior, and its posterior expectation, given the standardized one-step surprise z_k^2 (the squared residual of observation k against the agent's

own prediction, in units of its predictive variance), is the classic Student-t weight

$$\eta_k \;=\; \frac{\nu+1}{\nu+z_k^2}, \tag{6}$$

applied multiplicatively to the channel's evidence deposit: *trust what does not keep surprising you*. The single dial ν sets how quickly surprise becomes doubt, and $\nu \to \infty$ recovers the Gaussian defaults exactly. Pointed *between* agents, the same functional is precision learning at the social level, $\eta_{ij} = (\nu_s + 1)/(\nu_s + z_{ij}^2)$ on the calibrated disagreement between i and j—readable from *opinions* or from *wiring*, the contested coupling entries of Π (Appendix C). These weights add no objective and no memory, stay strictly positive, and default to off, reproducing the baseline byte-identically.

Weights over structures. This is where the diversity claim lives. Inferred reliability says how much to trust a source; it cannot represent the *alternative* a source argues for. So each agent carries K candidate wirings (Π_k, h_k) and a posterior over a model index—$q_i(x, m)$ rather than $q_i(x)$—each candidate scored every step by its prequential predictive density *before* the shared deposit lands, so structure-discriminating evidence accumulates in the log-weights L_{ik} instead of being erased. Communication changes accordingly: parameter beliefs fuse only *within* a frame (candidate k with neighbours' candidate k, never averaged across frames), and what travels between frames is the hypothesis itself, by damped log-linear pooling of the log-weights, $L \leftarrow (1-\alpha_m)L + \alpha_m WL$. The damping gives the pooled log-odds a fixed point at a constant gap set by each camp's own evidence ($\sim s/2\alpha_m$ nats for two symmetric camps), where reliability weights alone leave consensus as the only fixed point. $K = 1$ reduces exactly to the baseline's fuse-then-observe, so the comparisons of §4 run all three architectures on byte-identical observation streams (Appendix C).

3.4 The simulation at a glance

The sweep of Appendix B instantiates the phlogiston world: a hub-structured paradigm net in which one core commitment is load-bearing for combustion, calcination and respiration; at a fixed step the world flips (the calx is heavier—the gravimetric anomaly), and the truth thereafter contains a dimension (oxygen) in *no agent's network*. One round of the simulation, for every agent:

1. **Observe.** The world emits observations at the belt; disconfirming channels carry the conviction-gate weight w_k (§3.2) and, with the reliability gate on, the inferred η_k (Eq. 6).
2. **Infer.** Exact linear-Gaussian update of the value-tilted posterior (Eq. 3); precision accumulates.
3. **Crisis check (reduction).** Candidate prunes are scored by $\Delta F + \lambda_i \Delta U$, both terms closed-form from the Savage–Dickey reduced posterior (Eq. 5); both protective belt edges clearing the bar is a *crisis*, and the hub is released (Appendix D).

4. **Fuse.** Posteriors and structural edits pool across the trust graph, weighted by the intra/inter coupling and, with the social gate on, by η_{ij} (§3.3); pooling all dimensions vs. only those jointly represented is a protocol choice with consequences (Appendix B).
5. **Forget.** Accumulated precision decays by ω (§3.2).

Crisis and reduction are thus not stipulated events but threshold crossings of the model's own evidence; Appendix D derives the check from the single tilted objective and lists explicitly what remains a modelling commitment. Note what the loop does *not* contain: an expansion step. By the scope choice of §3.2, the model only ever prunes; novelty in the main text is supplied, never conceived, and only the earlier phlogiston prototype reported in Appendix B runs the full cycle with a Poisson-arrival expansion check. Table 1 (Appendix B) lists the dials swept; everything else is held fixed, and all code is seeded and ships its own null and positive controls.

One thing deserves emphasis before the results: every baseline ingredient above is its field's default—Kalman-filter belief updating, the standard formalization of motivated reasoning, classical opinion pooling in information form, textbook Bayesian model comparison. Nothing exotic has been smuggled in; the results test what these defaults can do *together*.

4 Results

All results are from the population of §3.1, run by the loop of §3.4; the full Kuhnian-cycle sweep on the phlogiston world is summarized, with the simulation described, in Appendix B, and experiment details are in Appendix C. The nets are hand-wired to expose the geometry, not fitted to data.

4.1 Averaging collapses structural diversity; represented rivals hold it

The baseline behaviour follows from the fusion step itself. Fusion is precision-weighted averaging on a connected trust graph—DeGroot dynamics in information form [55, 60]—and opinion dynamics has long known what such pooling does: it is a contraction, so disagreement decays geometrically toward consensus, and a Gaussian agent never rejects a conflicting source—it averages it in [4–6]. Active inference has begun to ask the same question of its own fusion step—whether belief sharing is a blessing or a curse [2]—and over structured Bayes nets the answer has a one-line algebraic face: $|\operatorname{mean}_i \Pi_i| \le \max_i |\Pi_i|$ entrywise, so an average can never *create* an edge absent from every individual—whatever a collective knows beyond its members lives in the *union* of their networks, never in their average.

The experiment: two camps in full contact on a shared world, each running only its own commitment's experiment—theory-laden evidence, the Kuhnian condition—and three architectures on byte-identical observation streams: plain

fusion (the default), the Student-t trust weights of §3.3, and the structure posterior of §3.3. We measure the inter-camp *structure distance* $D(t)$: small when the camps wire the world the same way, large when they hold different wirings.

Figure 3 is the result. Under plain fusion the inter-camp distance collapses within a few steps: the contraction. Under the trust weights the gap holds for hundreds of steps, then visibly decays—the weights are memoryless and stay strictly positive, so consensus remains the only fixed point; what they buy, they buy as time. (What the weights *read* also decides: pointed at opinions they never fire, because both camps fit the same blended data and agree in their means; pointed at the wiring they retain 94% of the divergence where fixed trust keeps 10%—Appendix C.) With represented rivals—K candidate wirings and a posterior over them, so each camp can see the difference between the models—the gap stays flat to $t = 1500$, each camp internally certain of its own theory: a fixed point of the damped hypothesis-pooling dynamics, not a slow leak (Appendix C).

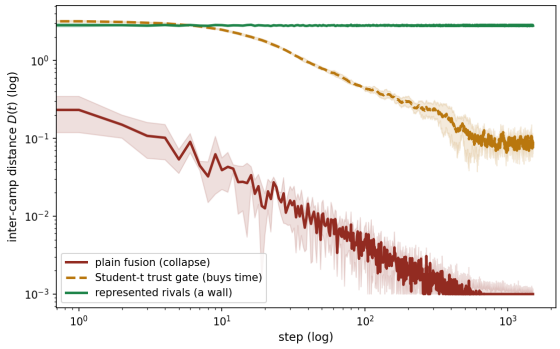


Fig. 3: Same setups, same evidence—only the representation differs (two camps, full contact, theory-laden streams; inter-camp distance $D(t)$, mean ± s.d., 5 seeds, log–log). Plain fusion collapses (the contraction); the Student-t gate decays (time, bought and spent); represented rivals hold. Each architecture runs its own best memory setting (Appendix C).

4.2 The Kuhn cycle with represented rivals: crisis as a state

With the structure posterior in place, the Kuhnian transition itself can be run (Fig. 4). A population holds two candidate wirings of the phlogiston world—the incumbent and the oxygen reading—and the world turns at the dotted line. Normal science forms endogenously: from a uniform start, the population commits to the incumbent and q(oxygen) sits at zero. After the world turns, the model posterior goes *torn*: the entropy of $q(m)$ rises to a peak while q(oxygen) climbs—crisis as a *state* of the belief, not a stipulated event—and then resolves as the

population flips to the new paradigm and recommits: revolution, then a new normal. In the Gaussian baseline none of these phases exists: there is no $q(m)$ to go torn, and the transition is invisible from the belief state.

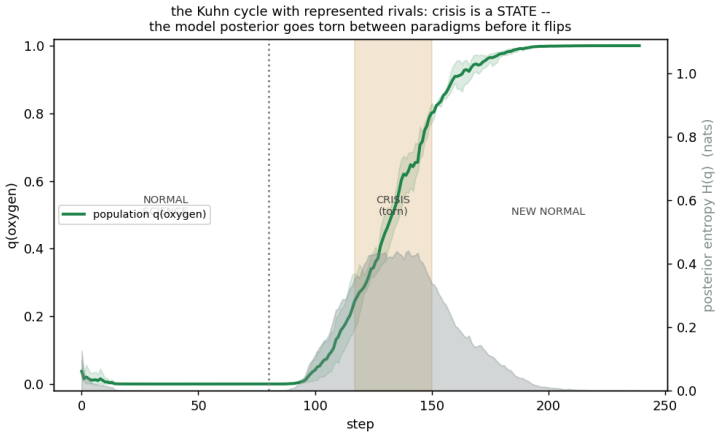


Fig. 4: The Kuhn cycle with represented rivals (population mean ± s.d.; the world turns at the dotted line). Green: the population posterior on the oxygen reading, q(oxygen). Grey: the entropy of the model posterior $H(q(m))$, peaking in the shaded band—crisis as a torn state between paradigms—before the flip completes. Normal science, crisis, revolution, new normal, all read off one belief state.

Boundaries, stated plainly. The demonstrations are deliberately small—one agent, $N = 60$, then $N = 16$, on hand-wired worlds—existence proofs that the persistence was representational, not sweeps of social topologies (the one such sweep—this transition replayed across four reference contact graphs—is in Appendix C). No model expansion runs anywhere in the main text (§3.2): the rival mechanism is *given* its K candidates, and anchored forgetting on the candidate nets is part of the mechanism—keeping a rival represented means not letting accumulated evidence drown it. What it does not yet do is named: under log-linear pooling a lone discoverer's candidate is majority-swamped, and no agent yet *acts* on where its candidates disagree—the crucial experiment. Controls: a world that never shifts produces no doubt; gates and rivals off reproduce the baseline byte-identically; no bar is tuned (Appendix C).

5 Discussion

Taken together, the results return us to the opening premise. Communities of reasoning agents—scientific ones, and increasingly the artificial ones built around

them—are asked to be *open-ended*: to keep representing new structure, not merely to refine the structure they already share. The results say what that requires. When each agent's paradigm is a single Gaussian belief net and combination is precision-weighted averaging, divergence does not survive contact: the community converges on a compromise network no member holds, however the dials are set (Result 1). When agents can instead hold different structures apart—candidate wirings with a posterior over them, hypotheses communicated rather than averaged—divergence survives full contact (Result 2). Stated as one finding: a community doing Bayesian model reduction over structured Bayes nets needs the ability to represent the *difference between models* inside each agent. Without that ability there is no divergence; without divergence, no open-endedness.

The homogenization is thus a failure of *representation*, not of tuning. There is no coordinate in (Π, h) whose value can mean *torn between two hypotheses*: conflict arrives as growing confidence, and fusion erases who disagreed about what—which is why no dial of the sweep escaped it, and why inferred reliability could only slow it. The structure posterior of §3.3 is that diagnosis made executable: multiple structural hypotheses per agent (an agent that can be torn), divergence between live candidates as a computable state, and communication of hypotheses—fusion within frames, never averaging across them.

The relation to Kuhn is now allegory-free: the regime *holds* normal science (the derived core and belt of §3.1). Of the revolutionary remainder, inferred reliability *supplies* crisis as suspended confidence and renders incommensurability [65] a timescale, while represented rivals make it a *state*: two camps internally certain of different wirings at a fixed point, the gestalt switch given its slot (a $q(m)$ that flips on prequential evidence). And any community whose aggregation step averages rather than gates inherits the regime's signature, whatever the sophistication of its members—the same sharing that lets a group synchronise its members' anticipations and act as one [46] is what averages the dissenter away: a population of structure learners is only as open-ended as its pooling permits—and as what its pooling *reads* (§4.1)—a property of the protocol, designed in rather than an emergent grace of scale.

Limitations and future work. The demonstrations are small and the nets handwired; the claims are existence proofs about representation, not sweeps of social topologies. The clearest next step is the expansion this paper deferred (§3.2)—and the structure posterior suggests where to point it. The divergence between an agent's live candidates is a computable map: the KL divergence between the predictive distributions their couplings induce, the same functional Bayesian model reduction already scores prunes with (Eq. 5). Read forward instead of backward, that map is a natural *prior over where to expand*: new structure is most likely to pay exactly where the represented models disagree. Beyond this—and the swamped discoverer and unchosen crucial experiment already named in §4.1—lies the deepest exit: a new *instrument* enlarges the likelihood's own support, and one cannot place a posterior over the unconceived [27]. What a community controls is how hard it looks—and, above all, what its pooling does to the member who finds it first.

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A The operator and the tilt: the closed forms behind the two fields

This appendix is the closed-form backing for the paper's propagation machinery: it states the structural equation model behind $p(x \mid G)$, why the operator $\mathsf{T} = (I - A)^{-1}$ is a finite sum, and what the value tilt of Eq. (3) does to the posterior, exactly. Consult it for the algebra behind the two fields $\kappa = \mathsf{T}\mathbf{1}$ (Eq. 1) and $U = \mathsf{T}u$ (Eq. 2), for the mind-change cost the revision objective prices (Eq. 8), and for the transpose convention needed to reproduce either field against the released code. Nothing here is required to follow the results; it is the warrant for the main text's claim that every quantity it uses is closed form.

How propagation is represented. The couplings form a structural equation model $x = Ax + \zeta$, $\zeta \sim \mathcal{N}(0, D^{-1})$ with A the weighted DAG adjacency [18]; the choice is for tractability, and a nonlinear coupling would replace T by a local linearization. The propagation operator is the resolvent

$$\mathsf{T} = (I - A)^{-1} = \sum_{k \geq 0} A^k, \qquad \mathsf{T}_{iw} = \sum_{\text{paths } i \rightsquigarrow w} \prod \pi \;=\; \pi_{i \rightsquigarrow w}, \tag{7}$$

the Neumann series finite because a DAG makes A nilpotent. We realise T on the coupling magnitudes $|A|$ so downstream mass does not cancel when a child opposes a parent; the signed A builds the precision matrix. We index T so that κ_i sums row i; under the SEM convention the released code stores the transpose and computes the same fields as column-sums of $\mathsf{T}^\top$, so a reader reproducing $\mathsf{T}\mathbf{1}$ literally against the code's array must transpose first.

The value tilt. With honest posterior $p(x \mid o, G) = \mathcal{N}(\mu, \Sigma)$, the tilted posterior of Eq. (3) is a pure mean shift, $q_\lambda(x) = \mathcal{N}(\mu + \lambda \Sigma U, \Sigma)$, with cumulant-generating function $Z(\lambda) = \lambda\, U^\top \mu + \frac{1}{2}\lambda^2\, U^\top \Sigma U$. Note that gating a channel ($\rho_k \to 0$) *raises* the posterior variance and hence Z''; lock-in lives in the vanishing mean-update gain $\rho_k H_k \Sigma U$, not in the curvature.

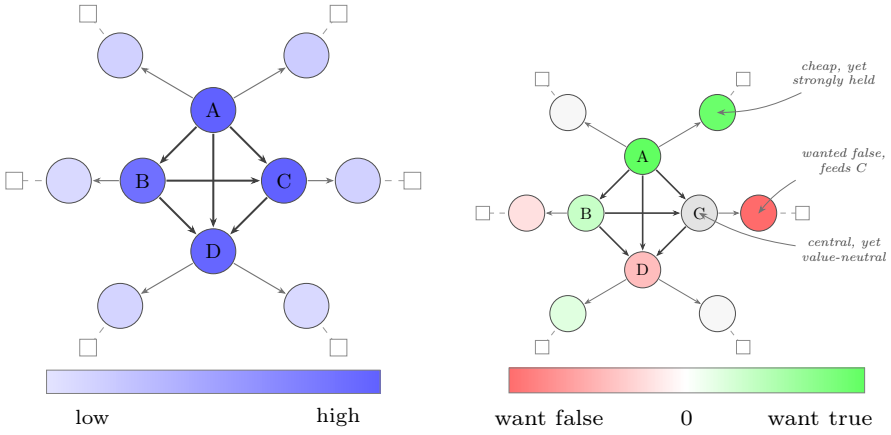
The mind-change cost of Eq. (3) is anchored to the network prior $p(x \mid G) = \mathcal{N}(\mu_0, \Pi_0^{-1})$: the revision q_λ pays

$$D_{\mathrm{KL}}[q_\lambda \,\|\, p(x \mid G)] = \tfrac{1}{2}\, \delta^\top \Pi_0\, \delta + \tfrac{1}{2}\big[\mathrm{tr}(\Pi_0 \Sigma) - \log\det(\Pi_0 \Sigma) - d\big], \tag{8}$$

with $\delta = \mu + \lambda \Sigma U - \mu_0$ and $d = \dim x$: the evidence-driven and value-driven displacements are priced *together*, in the prior precision Π_0—exactly the off-diagonal mass the carry-over field $\kappa = \mathsf{T}\mathbf{1}$ summarizes. The smaller quantity $D_{\mathrm{KL}}[q_\lambda \,\|\, p(x \mid o, G)] = \frac{1}{2}\lambda^2\, U^\top \Sigma U$ is the fidelity gap to the honest posterior, not the cost of changing one's mind: it would vanish for an agent that followed the evidence exactly, however far the evidence moved it.

B The null regime, measured

This appendix does two jobs for §4.1. First, it describes the simulation the main text only sketches: the phlogiston world—the paradigm network the paper's experiments instantiate—and the per-round community loop as actually run in the



(a) Revision cost κ_i (conservatism) (b) Intrinsic utility u_i (conviction)

Fig. 5: Two fields on one paradigm (schematic). Both panels carry the *same* network—a densely coupled core $\{A, B, C, D\}$ and a sparse belt $\{b_1, \ldots, b_6\}$ in contact with observation. a Revision cost is the carry-over of mind-change, $\kappa = \mathsf{T1}$ (Eq. 1): a node's cost is the precision-weighted mass of its descendants, a clean radial gradient monotone in centrality. b Intrinsic utility is signed—beliefs one wants true (green), wants false (red), or is neutral toward (white)—and the conviction field $U = \mathsf{T}u$ does not align with κ: C is central yet value-neutral; b_2 is cheap yet strongly held; b_4, which feeds C, is a conclusion the community wants false. The two fields are $\mathsf{T1}$ and $\mathsf{T}u$, the same operator on two sources, and they are independent vectors.

released experiments, in the paper's own notation. Second, it preserves, clearly flagged as such, the headline numbers the main text still cites from an *earlier and larger prototype* of the same world: a full Kuhnian cycle with a Poisson-arrival expansion step (Appendix D) and four closed-loop escape mechanisms, none of which the model of this paper contains (§3.2). Read it after Result 1, when you want to know exactly what was run and where each quoted number comes from.

The phlogiston world. Ten named commitments form the shared basis of every net: the hidden hub (*phlogiston*), eight surface commitments it explains—combustion and calcination as release, the metal as calx-plus-phlogiston, the sign of the mass change, gas consumption, the air's finite capacity, reduction with charcoal, respiration as slow combustion—and the anomaly node, *the calx is heavier than the metal.* The incumbent paradigm is a common-cause DAG: the hub is a parent of the eight surface commitments (which is what makes the phenomena co-vary), a belt mass-balance edge makes the anomaly a child of the two mass-law nodes, and the prior means get the anomaly *wrong*—phlogiston escapes, so the calx should be lighter. The rival paradigm drops the hub couplings entirely and keeps the belt mass law with the *correct* means: exactly the

direct couplings that marginalizing a hub leaves behind (the Schur residue, Appendix D). The world flips at a fixed step t_{shift}—the calx really is heavier thereafter. Observation is the gravimetric operator H: one direct row per measured commitment (the level signal) plus a single relational mass-balance row reading calx − mass-change − gas, the only row whose Fisher deposit $H^\top \operatorname{diag}(w)\, H/\sigma_o^2$ carries off-diagonal information—the row that can move structure.

The community and the round. The released experiments run the loop of §3.4 in its represented-rival configuration (§3.3; definitions in Appendix C): N agents on a contact graph, each carrying the K candidate wirings above (K=2, incumbent versus rival; a K=3 ladder adds the *articulated* incumbent—hub kept, mass means patched). Per round: hypotheses pool, $L \leftarrow (1-\alpha_m)L + \alpha_m WL$; parameter beliefs fuse within frames; every candidate is scored by its prequential predictive density before the shared deposit lands; the deposit lands in every candidate; anchored forgetting relaxes each candidate toward its own prior at rate ω. (K=1 is exactly the fuse-then-observe baseline—the null regime itself.) Theory-ladenness enters as attention: a conservative community carries weight $1-\gamma$ on the disagreement and mass-balance rows, where γ is precisely the silenced disconfirming fraction of Eq. (4)—dialled directly here rather than derived from the conviction field through $w_k = \exp(-g|U\cdot H_k|)$ as in §3.2—while a vanguard that keeps running the gravimetric experiment retains full attention. Runs are two-phase: a shared full-attention normal-science era to t_{shift}, then per-community attention. Neither structural check is in play in these runs by construction: the candidates are given, so nothing is pruned or proposed, and paradigm change is carried entirely by the model posterior $q(m)$. Defaults: N=24 with a vanguard of 4–8, t_{shift}=80, σ_o=1, ω=0.9, α_m=0.02 (the K=3 ladder runs σ_o=0.5, ω=0.5). The sweep is the one place expansion runs: a Poisson-arrival prototype of the discovery check, derived in Appendix D; no main-text result rests on it.

Only total isolation protects a superseded paradigm (earlier prototype). This sweep, run on the prototype with fixed trust, pits a dogmatic community—high conviction, whose agents gate their own disconfirming channels (§3.2)—against an open vanguard already tracking the anomaly, and dials the inter-community trust weight *inter* from 0 (sealed off) to dense contact. Sealed off, the gating protects: under strong gating ($s = 10$) only 19% ever convert. But at *inter*= 0.005, 92% convert to oxygen, however strong the gate and however dogmatic the agents, and the transition is a sharp boundary, not a gradual fade (Fig. 6). A control pins down what does the stalling: keep conviction exactly as high but switch the self-gating off, and the community revolts in full (revolted fraction 1.0 versus 0.0, $\gamma \to 1$). The tilt never stalls anyone (Eq. 4); its one causal route to lock-in runs through the gate, and the gate loses to the faintest connection, because a neighbour's evidence arrives weighted by trust, not by the receiver's conviction. Motivated reasoning in this regime is *wishful*, not *dogmatic*: a bias lives in the potential h; an entrenchment would have to live in the precision Π—which is exactly where the inferred-reliability gate of §3.3 puts it.

Dial	Symbol	What it controls
Contact	β	bridge density between the two communities (0 = isolation)
Topology	—	the contact graph: cycle, wheel, two communities, complete
Vanguard fraction	—	agents who keep running the disconfirming experiment
Self-censorship	γ	attention $1-\gamma$ on disconfirming rows (Eq. 4, dialled)
Hypothesis pooling	α_m	damping of the log-weight pool (§3.3)
Forgetting	$\omega \in (0,1]$	anchored decay of each candidate toward its own prior
Observation noise	σ_o	sensory noise at the belt
Candidates	K	wirings carried per agent (1 = the null baseline)
Earlier prototype only (the flagged paragraphs below):		
Social coupling	*inter*	inter-community trust weight (0 = isolation)
Conviction	λ_d	dogmatic community's value tilt / prune protection
Gate scaling	—	strength of the conviction gate on disconfirming channels
Proposal rate	r	Poisson arrival of structural proposals (Appendix D)
Reliability gates	ν, ν_s	Student-t gates on channels / fusion weights (off = Gaussian)
Fusion protocol	—	pool all dimensions (delta) vs. jointly represented (abstention)
Closed-loop mechanisms	—	adequacy-driven rate, social excitation, audit credit, value accretion

Table 1: The dials; everything else is held fixed. Above the rule, the dials of the released simulation described in this appendix—every gate and mechanism defaults to off, reproducing the fixed-reliability baseline byte-identically. Below the rule, the dials specific to the earlier prototype whose still-cited numbers follow, flagged paragraph by paragraph.

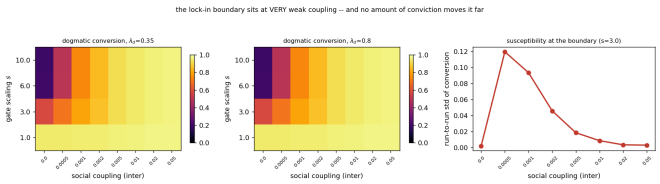


Fig. 6: Only isolation protects a superseded paradigm (phlogiston sweep, fixed trust). *Left, centre:* the fraction of the dogmatic community that converts to oxygen, as a function of contact (*inter*) and of how strongly conviction gates disconfirming channels (s), at two conviction levels ($\lambda_d = 0.35$ and 0.8). Past *inter* ≈ 0.005 nearly everyone converts, and doubling conviction barely changes the picture. *Right:* the run-to-run spread spikes in a narrow band around *inter* $\approx 5 \times 10^{-4}$: a sharp boundary, not a fade.

Pooling kills the lone discoverer—by protocol, not by principle (earlier prototype). In the prototype, discovery means waking a node—oxygen—that no agent's network yet contains; candidates arrive at a Poisson rate r and are accepted on the model log Bayes factor (Appendix D), so the fraction that discovers follows the waiting-time law. What the protocol governs is *survival*: delta pooling reads *has no such variable*—a fact about an agent's model class—as *maximally certain it is zero* [47, 62], so a discovery made by one agent at a time is averaged

away every round, and staggered survival is zero at every rate. Under dimension-aware *abstention* pooling—each precision entry pooled only over the neighbours that represent it—the survival threshold relocates wholesale: survival tracks the discovery fraction itself (0.39/0.64/0.86 at $r_0 = 0.005/0.01/0.02$; Fig. 7).

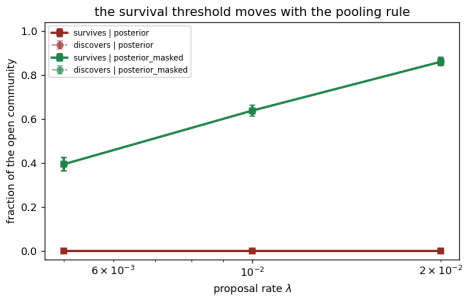


Fig. 7: The survival threshold moves with the pooling protocol (32 seeds per cell). Solid: fraction of the open community whose concept survives; dashed: fraction that discovers at all. Under delta pooling (red) a pinned slot is a confident "no such thing" and staggered survival is zero at every rate; under abstention pooling (green) survival tracks discovery. Incommensurability as a population-level force is a property of the protocol, not of the concepts.

Every escape goes around the averaging, never through it (earlier prototype). Four mechanisms were layered on the prototype baseline, and none lets two connected communities holding the same data *remain* in disagreement. An adequacy-inferred proposal rate—look harder when the residual is high—shortens crisis-to-discovery from 55 to 29 steps, matched by a constant rate of equal mean: it re-times the passage. Social excitation (a Hawkes kernel on neighbours' discoveries) collapses the waking dispersion from 64 to 2 steps and lifts staggered survival from 0 to 0.88–0.95: synchrony outruns the averaging rather than escaping it. An audited wake-then-prune cycle re-pins 100% of wakes in the null world: curiosity kept honest, nothing else changed. Value accretion—conviction grown from inside by each confirmation—delays conversion $\sim$2.2$\times$ and never prevents it: time bought, not immunity. Each either re-times the passage through the consensus bottleneck or synchronizes the population for it.

Robustness and controls (earlier prototype). The arc is robust across forgetting and observation noise ($\omega \in [0.9, 1.0]$, $\sigma_o \in [0.25, 1.0]$): crisis and reduction survive everywhere—including the no-forgetting wall $\omega = 1$, where only discovery thins: a community that never forgets still sees its paradigm die, but loses the capacity to grow the successor. Each experiment ships its own null and positive controls—a world that never shifts produces neither crisis nor discovery; the gated, isolated

community stays locked; Poisson discovery follows the waiting-time law—and no acceptance bar is tuned anywhere: the wake's sole accept test is the model log Bayes factor (Appendix D); whatever the derivations there do not fix is named there as a commitment.

C Inferred reliability and represented rivals: definitions and experiment details

This appendix carries the exact definitions of the two mechanisms Result 2 (§4.1) turns on—the Student-t reliability gate in its three reads (channels, opinions, wiring) and the represented-rival architecture (K candidate nets, prequential scoring, hypothesis-aligned fusion)—and then walks through every gated experiment behind the quoted numbers: what was set up, what to look at, and what each headline figure means. Consult it whenever a main-text claim cites a gate or a wall; the closing paragraph states the three commitments that bound all of it.

The three reads of one functional. Every gate in the paper is the Student-t scale-mixture weight $\eta = (\nu + 1)/(\nu + z^2)$—the posterior expectation of a source's latent noise scale given its standardized squared surprise z^2 (one E-step of the EM view of t-regression)—applied to three different surprises. *Channels:* $z_k^2 = (o_k - (H\mu)_k)^2/\mathrm{predvar}_k$ with $\mathrm{predvar}_k = \sigma_o^2 + (H\Pi^{-1}H^\top)_{kk}$, the agent's own surprise at observation k before depositing it; η_k multiplies the channel's deposit, composing with the conviction gate of §3.2. *Opinions:* $z_{ij}^2 = \sum_v(\mu_j[v] - \mu_i[v])^2/(\Pi_i[v,v]^{-1} + \Pi_j[v,v]^{-1})$, the calibrated disagreement of two posteriors summed over the measured nodes (summed, not averaged: a localized heresy must be able to sever trust); η_{ij} multiplies the fusion weights, whose row normalization absorbs the overall scale. *Wiring:* the same sum taken over contested coupling entries $x = \Pi[a,c]$, normalized by their pairwise symmetric scale $\max((x_i^2 + x_j^2)/2, \epsilon^2)$—bounded per coupling, and invariant to the global precision growth the forgetting equilibrium produces, so it reads *how* two agents wire the world, not how much evidence they hold.

Represented rivals: definitions. Each agent carries K candidate nets (Π_{ik}, h_{ik}) and log-weights L_{ik} over a model index. Per step, every candidate is scored by its prequential predictive density *before* the shared deposit lands, $\log p(o_t \mid o_{<t}, M_k)$, so structure-discriminating evidence accumulates in L rather than being erased by the Fisher deposit (which never sees the observation's value). Fusion is hypothesis-aligned—candidate k pools only with neighbours' candidate k—and the log-weights pool log-linearly with damping α_m: $L \leftarrow (1 - \alpha_m)L + \alpha_m WL$. Deviations of the pooled log-odds from the population mean converge to $(I - W)^+(s - \bar{s})/\alpha_m$, a constant gap of $\sim s/2\alpha_m$ nats for two symmetric camps: a genuine fixed point—the wall of Fig. 3—whose height the early-measured per-step source predicts conservatively (realized 110 nats at the horizon, tail still growing at +0.016 nats/step). Anchored forgetting on the candidate nets,

$\Pi_{ik} \leftarrow \Pi_{0k} + \omega(\Pi_{ik} - \Pi_{0k})$, is part of the mechanism: without it every candidate's precision grows without bound and their predictive densities converge. $K = 1$ reduces exactly to plain fuse-then-observe, which is how the Gaussian baselines run on byte-identical observation streams; each architecture runs its own best memory setting.

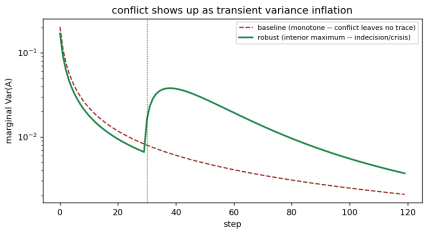


Fig. 8: Conflict as suspended confidence (§4.1). A channel turns contradictory at step 30. The Gaussian baseline (dashed) grows monotonically more certain—the Gaussian confident compromise; the gated agent's variance (solid) shows an interior maximum: doubt rises while data accumulate, then resolves.

The experiments, interpreted. All runs are seeded, assertion-checked at run time, and write their summary statistics next to their figures. The main-text experiments first. *Conflict crisis* (§4.1): one agent on a three-commitment chain whose channel A turns contradictory at t=30 (ν=4, T=120). Watch the posterior variance (Fig. 8): the gate infers the channel unreliable ($\eta_A \to 0.16$) and the variance peaks at t=38—doubt rising while data accumulate, the interior maximum no Gaussian can produce—while the violated edge carries 5.3× the strain of its neighbours, so the agent also knows *where* it is torn. *Divergent wiring* (§4.1): N=60 in a structurally underdetermined world, two attention camps, five seeds. The number to read is surviving cross-community structure: fixed trust retains 10% of the disconnected benchmark, the opinion-reading gate 10.2% (the camps agree in means, so it never fires), the wiring-reading gate 94%—what matters is *which* disagreement a gate can see, not that there is one. *Rival lock-on* (§3.3): one agent, K=2 wirings differing in nothing but structure (T=400): the prequential log-weights separate at 0.13 nats/step and $q(M_1) \to 1.0$, while the Gaussian pool's half-strength blur stays frozen ($< 10^{-3}$ drift)—the rival layer decides a question the pooled net cannot ask. *The wall* (§4.1): N=16, two camps, full contact, theory-laden channels (T=1500, five seeds): end-of-run disagreement 2.5×10^{-4} plain, 0.075 gated, 2.79 rival, with $q_{\text{own}} = 1.0$ in both camps—the fixed-point gap of Fig. 3, held, not decaying. *Targeted reduction*: one agent, a chain with one world-violated coupling (T=80): the violated edge is top-strain at every step and the ΔF verdicts are correct from step 1—reduction points where the world disagrees.

Population-level readings of the same gates. Four further runs put the gates through the full Kuhnian cycle; each is a further reading of the same two results—homogenization delayed, never escaped—and is reported here rather than in the main text. *Crisis trace* (N=20, world flips at t=40, ν=2, T=800): the community first *denies*—mean channel weight $\bar{\eta}$ falls from 0.85 to 0.17 and conversion stalls 4.3×—then *rehabilitates* the doubted channel ($\bar{\eta} \to 1.27$) once the revised wiring fits it, agreement channels staying > 0.85 throughout: denial and rehabilitation as one gate's trajectory. *Gated camps* (N=20, complete graph, opposed camps, ν_s=0.5, T=80): uniform fusion preserves 0.1% of initial disagreement, the gated run 48%, cross-camp trust collapsing to 0.8% of within-camp—self-organized isolation. *Schism threshold* (same setup, sweeping prior confidence at contact): fixed trust ends $\leq 0.2\%$ everywhere; the gate is inert below a confidence threshold ($< 1\%$) and holds 17/48/76% above it—schism feeds on accumulated conviction. *Lock-in rerun* (N=80, full cycle, ν_s=0.5, T=180, *inter* swept): conversion half-time stretches 78 → 146–170 and conversion drops to 0.54–0.73 versus 0.92 under fixed trust, because cross-community trust first collapses ($< 1\%$) and is later restored ($> 90\%$): isolation self-organized, then self-undone—delay without escape.

The transition's shape is the wiring's signature. A final run replays the Kuhnian transition of §4.2 where it must *spread*—N=24, a vanguard of 4 keeps full attention on the disconfirming experiment, the rest self-censor it (γ=0.99, a trickle)—on the four reference contact graphs (Table 1; 50 seeds). Every community completes the transition [32], but the *same* world flip arrives as four different events (Fig. 9): the median conservative converts at t=220/232/453/488 on complete/wheel/community/cycle, inter-agent spread 4/11/57/102 steps. The wheel keeps pace with the complete graph despite a λ_2 an order of magnitude smaller—the vanguard holds its hub: centrality, not density. The shape of a paradigm shift is a property of the contact graph.

Scope, stated plainly. Three commitments bound the results. (i) Both gates are *memoryless*: η_k and η_{ij} are instantaneous functions of current surprise, so every school and schism reported is metastability relative to a horizon, not a new fixed point—the contraction theorem owns $t \to \infty$, and the lock-in rerun shows the trade directly: a harsher gate (ν_s=0.2) holds the boundary longer but leaves its own undoing unfinished at the same horizon, because the lock and the merge run on the same clock. (ii) The schism control is normalized by the pre-contact stance gap, since one uniform fusion step collapses the fixed-trust camps before the first logged sample. (iii) Structure-as-precision-pattern is attention-driven in the linear-Gaussian class (the Fisher deposit $H^\top \mathrm{diag}(w) H/\sigma^2$ never sees the observation's value): a camp committed to a wiring the data do not support consolidates it exactly as strongly as the supported camps do, so the wiring gate can protect a pluralism but cannot tell a supported school from a merely committed one—the data's verdict on an unsupported theory lives in the means, and in the strain and reliability reads that live on them. This refuted our own prediction, and it is the right caveat on every wiring-gate claim.

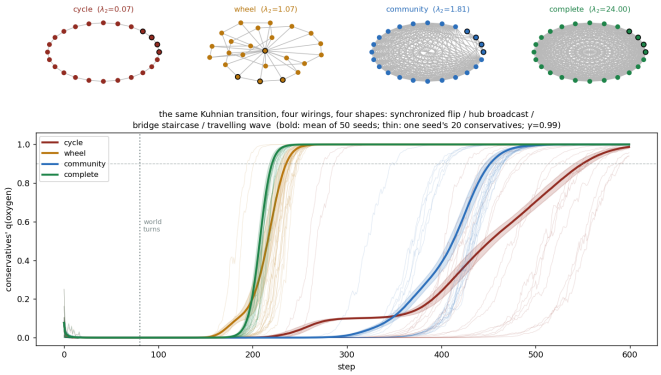


Fig. 9: The same Kuhnian transition on four contact graphs, on one canvas. Bold: conservatives' mean q(oxygen) (50 seeds ± s.d.); thin: one seed's 20 conservative agents; the world turns at the dotted line; vanguard nodes are outlined in the glyphs. Complete and wheel flip together and early; the community graph converts as a staircase after the bridge delay; the cycle as a travelling wave.

D The crisis check, the Schur residue, and the prototype's discovery check

This appendix derives the simulation's threshold events from the single objective the paper already carries, so that no structural change reads as a stipulated event. The *crisis check*—step 3 of the loop in §3.4, the only structural move the model of this paper makes—is derived in full. The *Schur residue* comes next: what marginalizing a hub leaves behind, the algebra behind both the oxygen rival of Appendix B and the discovery direction. The *discovery check* belongs only to the earlier prototype of Appendix B (the main-text model runs reduction only, §3.2) and is derived in compressed form so the prototype's flagged numbers remain auditable; whatever is *not* derived is named as a commitment at the end.

D.1 One ledger for every structural move

The bounded agent of §3.2 revises its beliefs by minimizing $D_{\mathrm{KL}}[q \,\|\, p(x \mid o, G)] - \lambda\, \mathbb{E}_q\big[U^\top x\big]$ (Eq. 3): fidelity to the honest posterior, traded against the value of where the commitments land, at temperature λ. A structural move—pruning an edge, waking a node—changes both terms, so the same objective scores the move:

$$\text{score}(M \to \tilde{M}) \;=\; \underbrace{\Delta F}_{\text{log Bayes factor}} \;+\; \underbrace{\Delta G}_{\text{epistemic gain}} \;+\; \lambda\, \underbrace{\Delta U}_{\text{conviction value}}, \qquad \text{accept} \iff \text{score} > 0, \tag{9}$$

with ΔF the evidence for the edited structure, ΔG the expected information gain about any newly introduced latent (zero for a prune), and $\Delta U = \mathbb{E}_{q_{\tilde{M}}}\big[U^\top x\big] -$

$\mathbb{E}_{q_M}\left[U^\top x\right]$ the change in the conviction-field value of the commitments. Equation (9) is not a new mechanism: it is Eq. (3) evaluated at two structures and differenced. Both threshold events below are instances of it.

D.2 The crisis check is the ledger on a prune

For a prune, every term of Eq. (9) is closed-form in the linear-Gaussian setting. The evidence half is Bayesian model reduction (Eq. 5): writing the agent's accumulated likelihood deposit as $J = \Pi - \Pi_0$, $j = h - h_0$, the Savage–Dickey identity scores the swap of the prior (Π_0, h_0) for the edge-e-reduced prior $(\Pi_0^{(e)}, h_0^{(e)})$ under the same likelihood, and—this is the part the main text does not spell out—it *already exhibits the reduced posterior*,

$$\Pi^{(e)} = \Pi_0^{(e)} + J, \qquad h^{(e)} = h_0^{(e)} + j, \tag{10}$$

so the value half costs one linear solve per candidate edge:

$$\Delta U_e \;=\; U^\top\big(\mu^{(e)} - \mu\big), \qquad \mu^{(e)} = \big(\Pi^{(e)}\big)^{-1} h^{(e)}, \quad \mu = \Pi^{-1} h, \tag{11}$$

with $U = \mathsf{T}u$ the agent's *current* conviction field, read along its present couplings. The crisis check is then exactly the ledger: edge e is flagged iff $\Delta F_e + \lambda\, \Delta U_e > 0$, and *crisis* is the round at which both protective belt edges are flagged, upon which the agent adopts the reduced prior composed over every edge its own ledger flags (the hub released). Pruning a conviction-protective edge moves the posterior mean away from the cherished commitments, so $\Delta U_e < 0$ and the rule reproduces the intuitive form "the evidence must overcome λ times the value the edge protects."

A natural surrogate suggests itself here: the static threshold $\Delta F_e > \lambda\, v_e$ with $v_e = |U_{\text{parent}}| + |U_{\text{child}}|$, the summed conviction magnitude at the edge's endpoints—available without a solve, and with the right *form* (it is Eq. (9) with $\Delta U_e \equiv -v_e$). But it has the wrong *content*: v_e is a static stand-in for a quantity the model can compute exactly, so we compute it. The difference is not cosmetic, and its direction is itself informative: because the reduced and full posteriors share the likelihood deposit, the mean displacement of a prune—and with it $\lambda\, \Delta U_e$—*shrinks as evidence accumulates.* A data-rich paradigm is barely protected at the threshold; almost all of its persistence must come from upstream, from the conviction gate that decides which evidence is allowed to accumulate at all (Eq. 4). The static surrogate would quietly overstate threshold protection; the exact ledger locates lock-in where the paper's results put it: gating plus isolation, not a high bar at the moment of decision.

D.3 The Schur residue: what marginalizing a hub leaves behind

Marginalizing a hub condenses its influence into new edges among its neighbours: for a common-cause structure over (x_1, x_2, b),

$$\Pi = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow{\text{marginalize } b} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \tfrac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{pmatrix}, \tag{12}$$

a direct coupling induced where the joint had a zero. Conditioning on a shared *effect* is the dual operation: two commitments that jointly feed one observation are a priori independent, and clamping that observation anti-couples them—explaining away, run on the precision matrix. Both operations fill in the off-diagonal coupling the carry-over cost $\kappa = \mathsf{T1}$ is built from, and Fig. 10 draws both on the smallest nets that exhibit them.

Two of the paper's objects are instances of this residue. The oxygen rival of Appendix B keeps exactly the direct mass-law couplings that marginalizing the phlogiston hub leaves behind—the incumbent's Schur complement over its hub, means corrected. Read in reverse, the residue is a *signature*: a hidden hub b with couplings c leaves, on marginalization, the rank-one footprint $-cc^{\top}/\Pi_{bb}$ in the visible precision—exactly what the prototype's discovery check reads off the residual below.

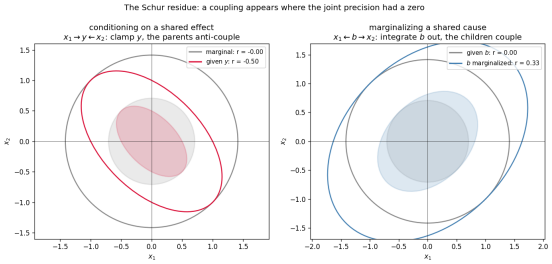


Fig. 10: The Schur residue, computed on the smallest nets that show it. *Left:* conditioning on a shared effect ($x_1 \to y \leftarrow x_2$): the parents are a priori independent (grey, $r = 0$); clamping their common descendant y manufactures an anti-correlation (crimson, $r = -0.50$)—explaining away. *Right:* marginalizing a shared cause (the matrix of Eq. 12): the children are independent given b (grey); integrating b out leaves them coupled (blue, $r = +0.33$). In both cases a coupling appears where the joint precision had a zero—the off-diagonal mass the carry-over cost $\kappa = \mathsf{T1}$ is built from.

D.4 The discovery check is the ledger on an expansion (prototype only)

This check runs only in the earlier phlogiston prototype of Appendix B; the model of the main text runs reduction only (§3.2). Expansion cannot be closed-form: a new node needs a real inversion, and—Stanford's problem of unconceived alternatives—the agent cannot plan toward a dimension it has not conceived. The check therefore has exactly two parts, and only two:

1. **Arrival.** Candidate proposals arrive at a Poisson rate r. This is the one place a stipulated process enters, and it is a deliberate modelling commitment: a rate is the maximally agnostic model of conceiving what one has

not conceived. The proposal's *direction* is read from the model's own prediction errors: by the rank-one footprint above, the leading eigenvector of the windowed error second moment is the best single-hub explanation of the residual. The window is an estimator horizon, not a gate.

2. **Acceptance.** The arrived proposal is accepted iff the model log Bayes factor of the bordered (hub-wired) model against the agent's current one is positive—iff the residual genuinely loads on the proposed hub. The epistemic gain ΔG sits on the other side of the move: being positive for *any* new latent, it pays for *attempting* the inversion, and it cannot keep a hub the data do not hold—an accept test that included it would wake on pure noise. Entertaining is paid by curiosity; keeping is paid by evidence. Repeated looks give the Bayes factor a small false-discovery rate, owned rather than hidden; its principled remedy is the audited wake-then-prune cycle of Appendix B, not a reinstated threshold.

A tempting third part is deliberately absent: a *trigger*, requiring the residual floor (the eigenvalue above) to clear some height before the Bayes factor is consulted. It would duplicate the accept test's job with a free constant, so the check has no tuned threshold: the rate carries Stanford's commitment, the eigenvector the direction, the ledger decides. (The floor is recorded as telemetry: the visible signature of the residual the old paradigm had been absorbing.)

D.5 What remains a modelling commitment

The derivations leave a short list of choices not forced by the formalism; the claim that "crisis, revolution and discovery are threshold crossings of the model's own evidence" is honest only with this boundary drawn:

- **The belt conjunction.** "Crisis" is operationalized as *both* named mass-law edges flagged in the same round. The per-edge accept is derived; the conjunction over those two edges is our operational definition of having eliminated the phlogiston mass-law reading.
- **The ordering — a result, not a commitment.** The prototype attempts expansion only after an agent's own crisis, but the ordering is enforced by the evidence rather than by fiat: running the identical proposal + ledger pipeline for every pre-crisis agent at every step (without applying it), the ledger rejects 13,180 of 13,185 pre-crisis agent-steps, and the five crossings are marginal ($\Delta F < 0.005$)—the repeated-look false-positive rate owned above. The intact paradigm absorbs the residual; there is nothing for the check to find until reduction exposes it.
- **The Poisson rate** r (swept), the prediction-error *window* (an estimator horizon), a short *warm-up* before reduction is permitted (an initialization guard), and the released hub's self-precision are quantitative choices of the same standing as the other prototype dials of Table 1.